# Multi-Faceted Customer Intelligence

## An End-to-End Data Mining Pipeline on Online Retail II

Data Mining 2026 — Final Project

**Team 4 — University of Information Technology, VNU-HCM**

Nguyen Minh Cuong (22520177) • Vo Thanh Danh (22520201) • Nguyen Vinh Dat (22520228) • Nguyen Huu Dinh (22520251)

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# The Problem

*Three siloed questions every retailer asks*

- WHO are my customers? — segmentation
- WHO will leave? — churn prediction
- WHAT do they buy together? — basket analysis

Solved in isolation, these analyses miss the joint signal that drives retention ROI.

**Why this matters**

**+5%**

customer retention

drives 25–95% profit growth
(Bain & Co.)

# Dataset — UCI Online Retail II

*UK-centric online retailer, 2009-12 to 2011-12*

| | | |
|---|---|---|
| **1.07M** raw transactions | **805K** after cleaning | **5,878** unique customers |
| **4,631** stock codes | **41** countries | **2 yr** time span |

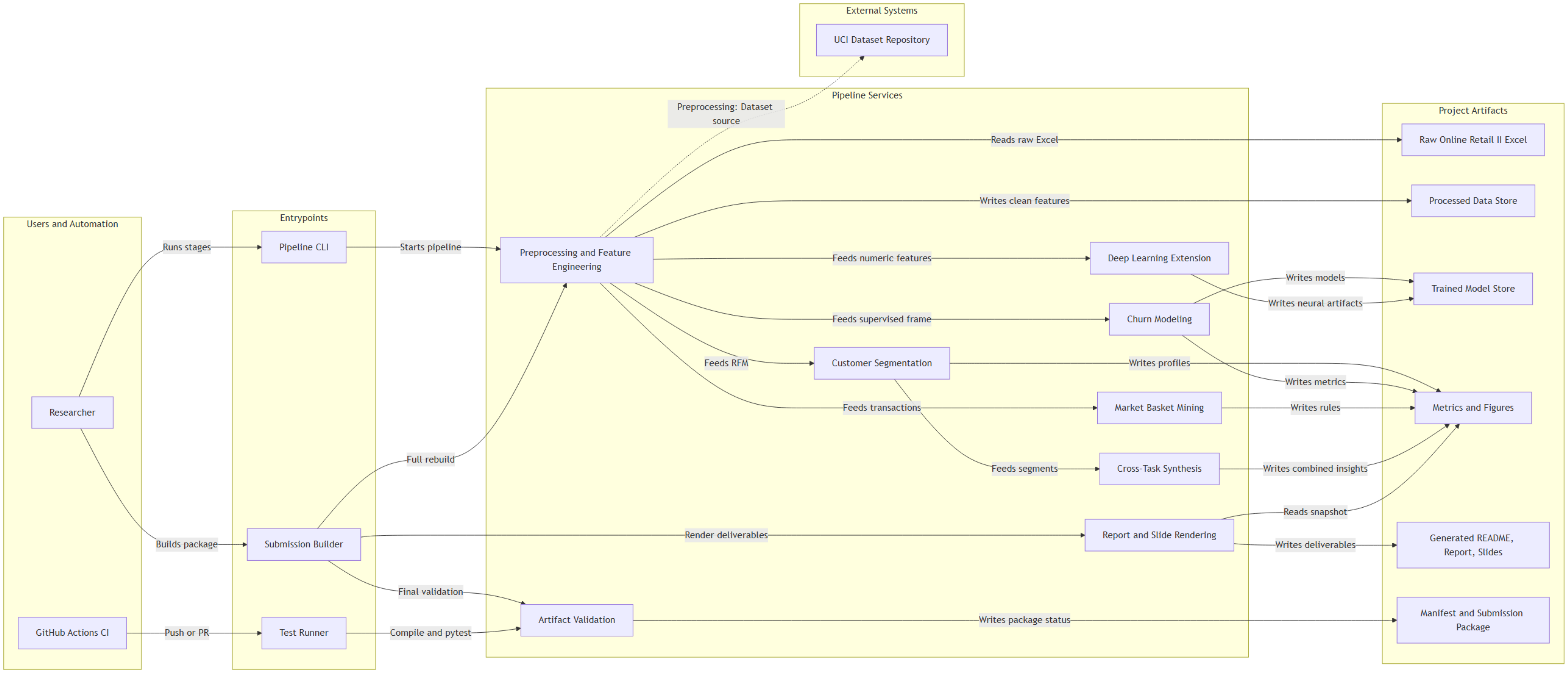
Columns (8, all transaction-level): Invoice · StockCode · Description · Quantity · Price · InvoiceDate · CustomerID · Country

**No customer attributes (name / contact / demographics) exist beyond the CustomerID key — a null CustomerID therefore carries no recoverable customer information.**

Source: https://archive.ics.uci.edu/dataset/502/online+retail+ii

# System Architecture

*Users & automation → entry points → pipeline services → artifact stores*



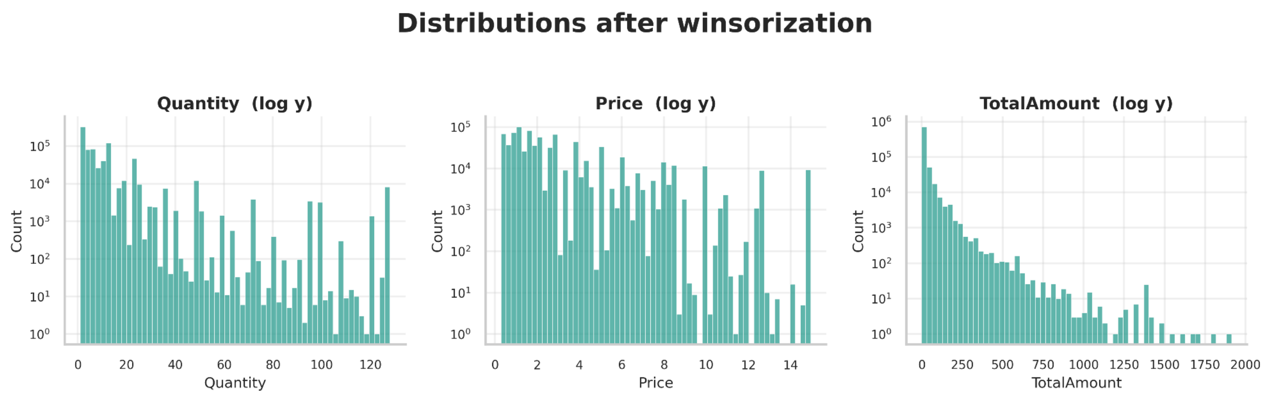
# Preprocessing

*From raw 1.07M rows to clean 805K rows — only 24.5% dropped*

- Drop 22.8% rows with null CustomerID
- Remove cancellation invoices (prefix 'C')
- Drop non-positive Quantity / Price
- Winsorize Quantity & Price at 1%/99% quantiles
- Derive TotalAmount = Quantity × Price
- Compute RFM + behavioral features per customer
- Label churn: no activity in 90 d after 2011-09-01

| Filter removed | % of raw rows |
|---|---|
| Missing CustomerID | 22.8% |
| Cancellation invoices (C) | 1.8% |
| Non-positive Qty / Price | 2.4% |
| **Total rows removed** | **24.5%** |

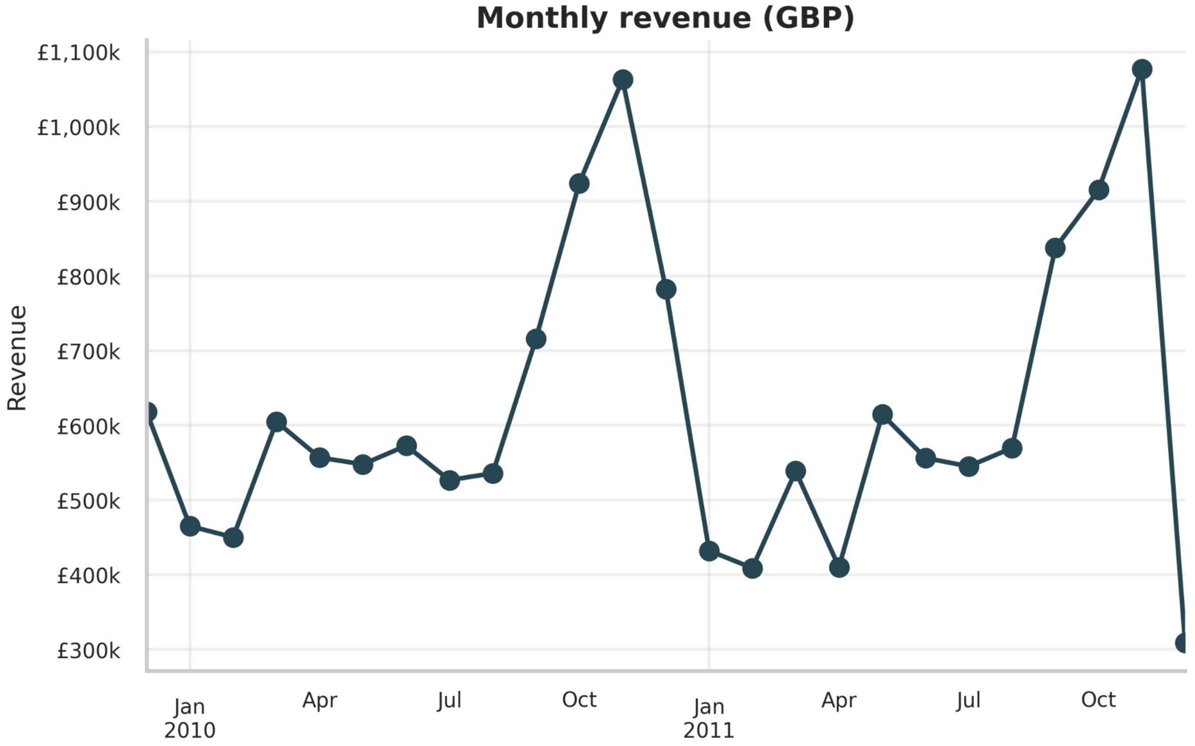
**Dropping null-CustomerID rows is mandatory (RFM / churn are per-customer) yet costs only 3.5% of units & 13.7% of revenue — almost no purchasing signal lost.**

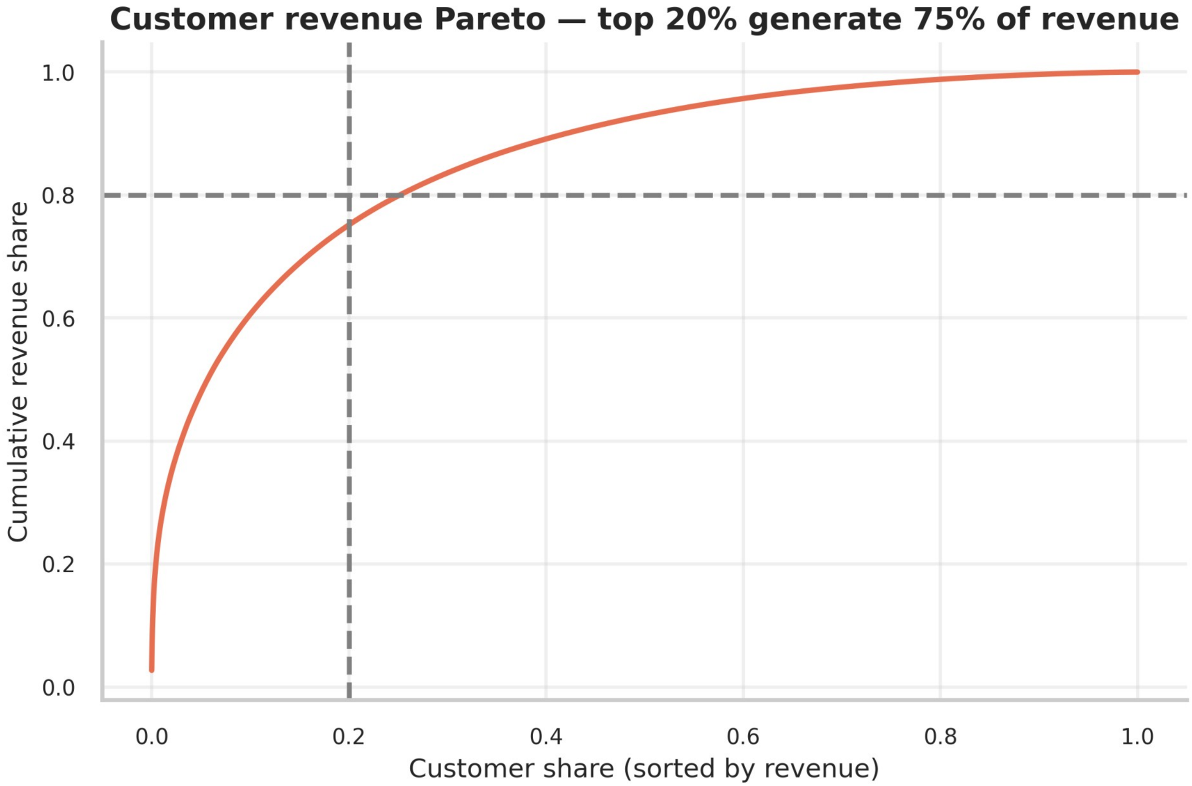



Quantity / Price / TotalAmount distributions after winsorization

# EDA — Revenue & Customer Pareto

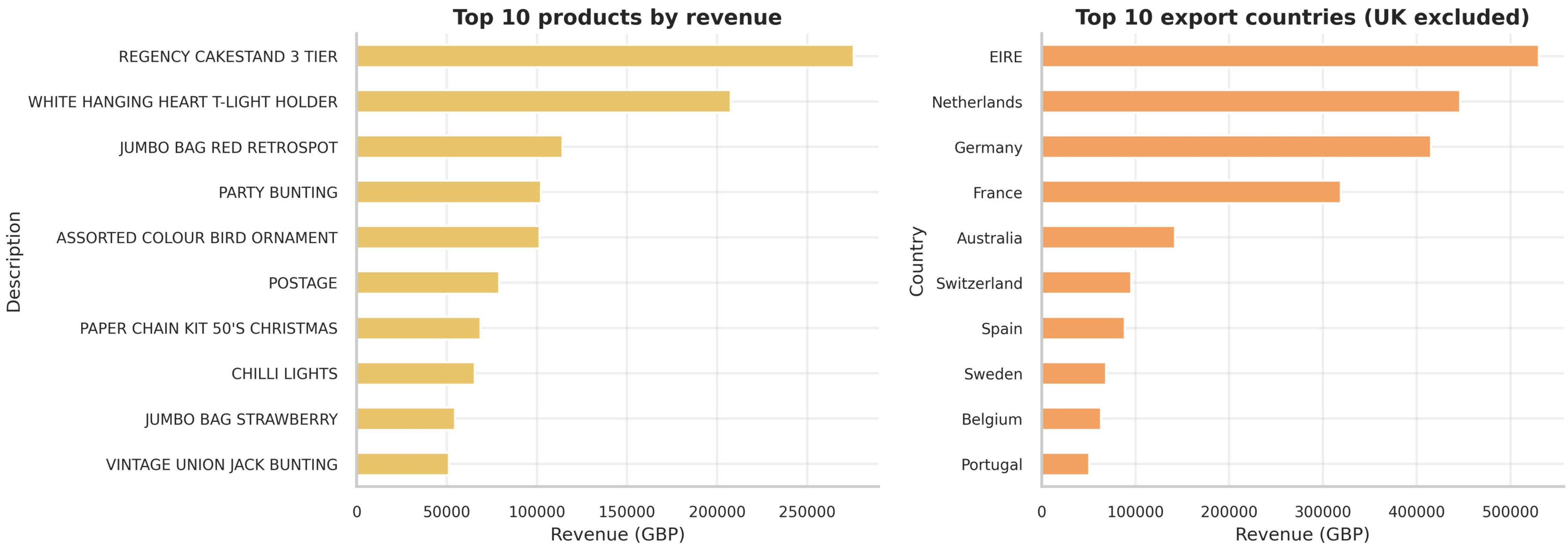
*Q4 seasonality + top 20% of customers drive most revenue*



# EDA — Top Products & Export Countries

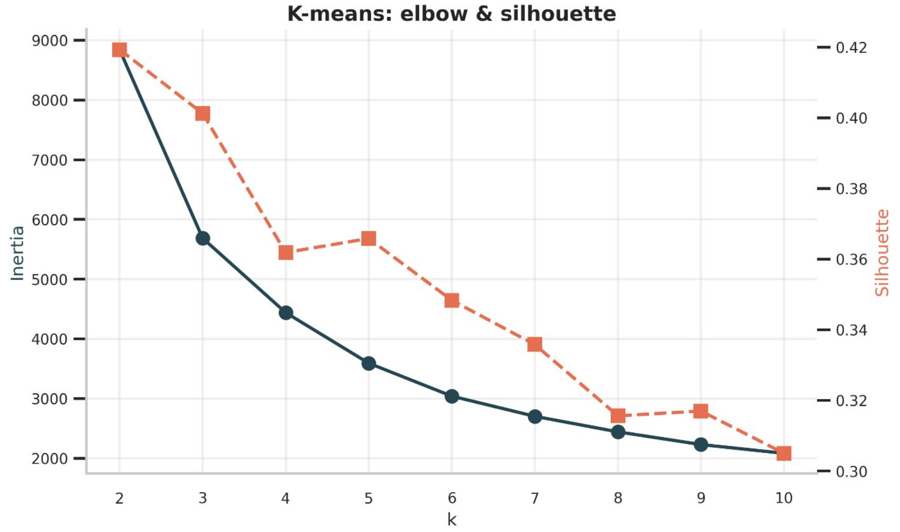
*Long-tail product mix; concentrated international demand*



# Customer Segmentation — Three Algorithms Compared

*K-means vs DBSCAN vs AGNES on log-transformed RFM*

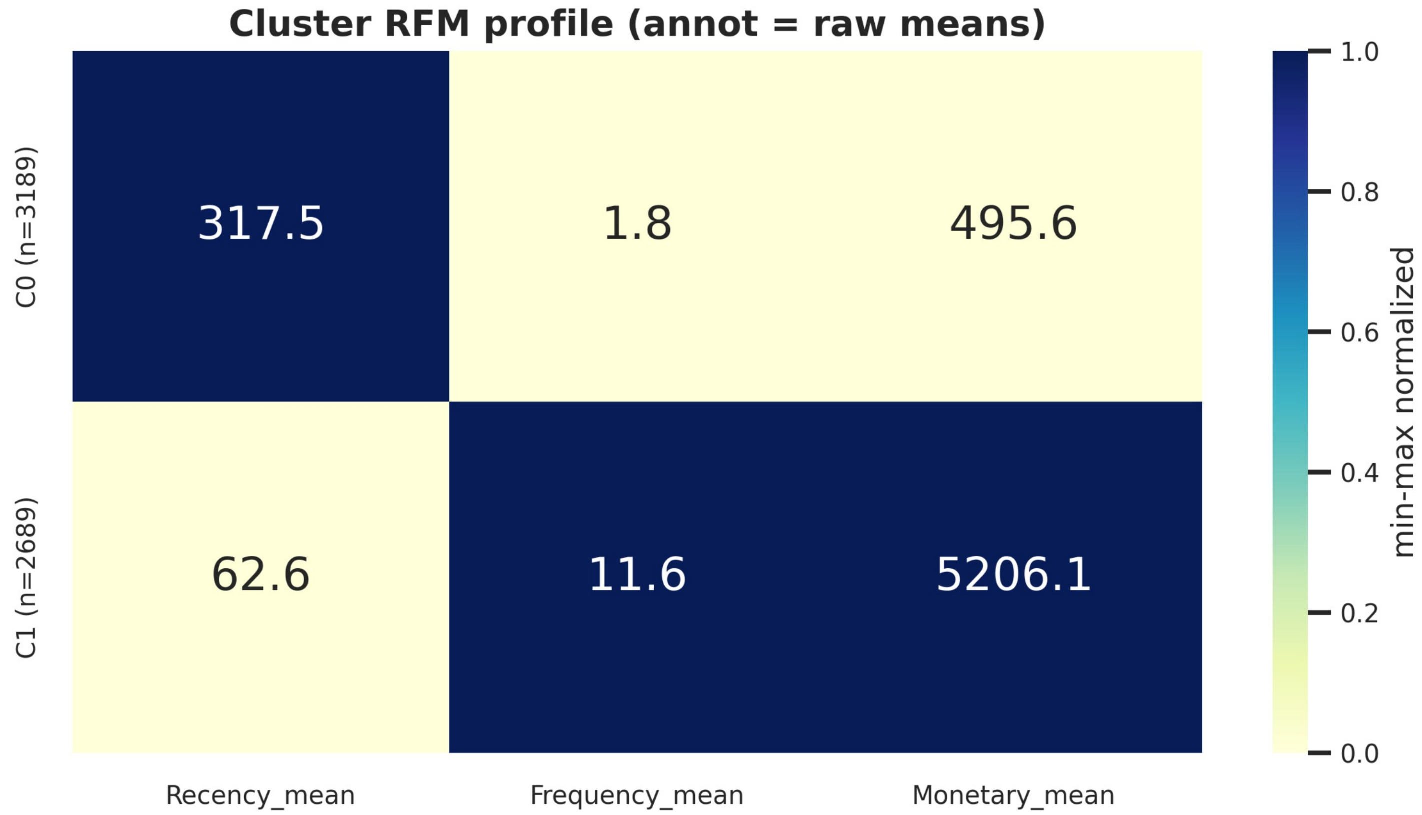
| Algorithm | k | Silhouette ↑ | Davies-B ↓ | Calinski-H ↑ |
|---|---|---|---|---|
| KMeans | 2 | 0.419 | 0.886 | 5840 |
| DBSCAN | 13 | -0.120 | 1.632 | 742 |
| AGNES | 2 | 0.388 | 0.789 | 4304 |




- K-means k=2 wins silhouette (0.42) — best partitional fit
- DBSCAN fragments into 13 + noise — wrong fit for heavy-tail
- AGNES Ward close second; chosen k=2 confirmed by dendrogram

# Two Customer Segments — Crystal Clear Interpretation

*Dormant (54%) vs Active loyal (46%)*




### Dormant

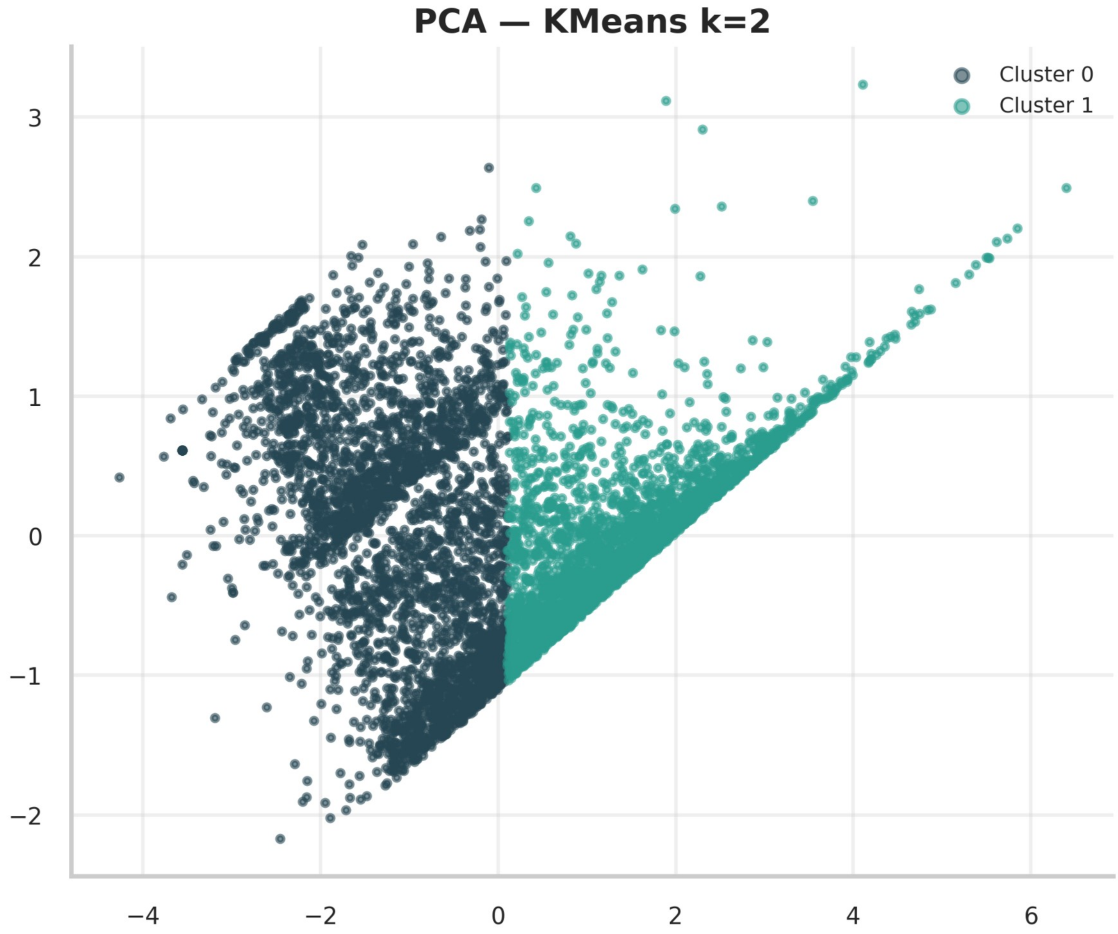
Recency 317 d
Frequency 1.85
Monetary £496

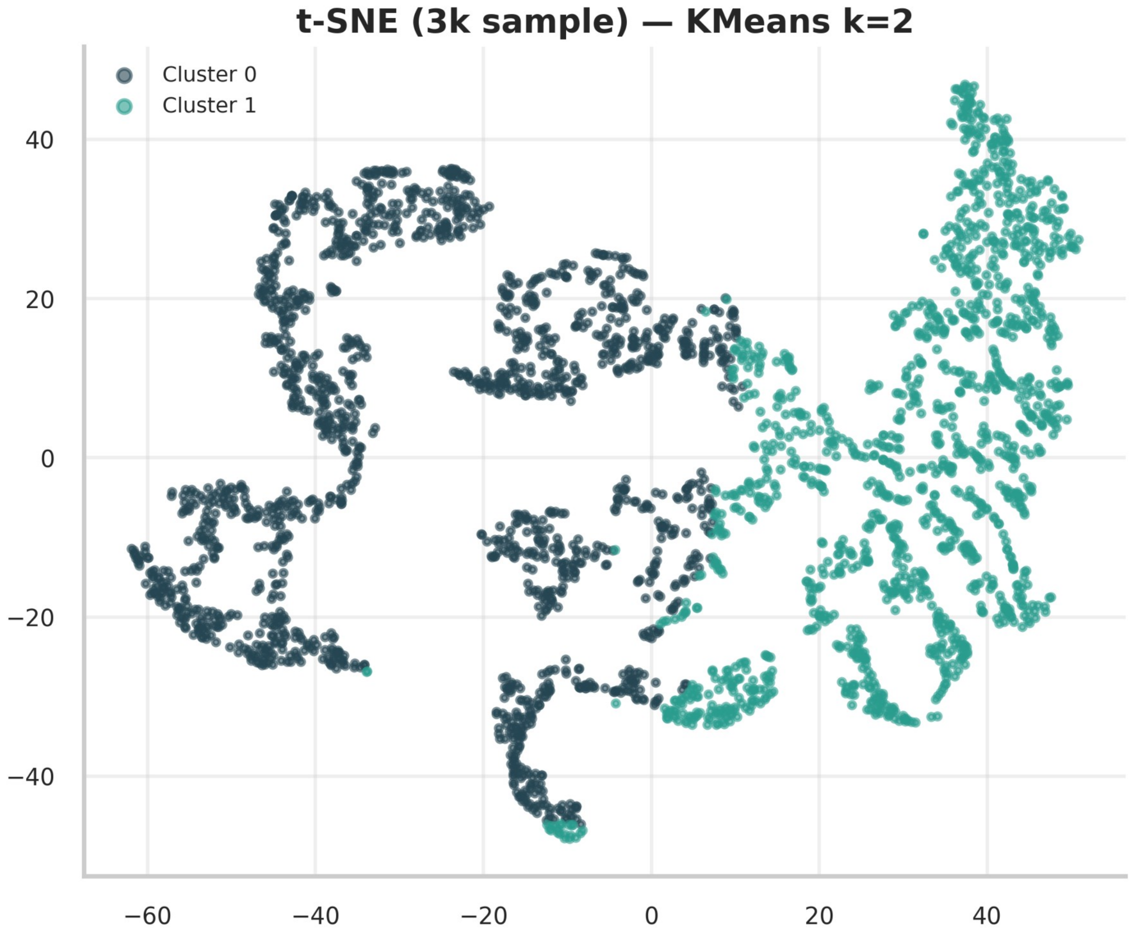
### Active loyal

Recency 63 d
Frequency 11.55
Monetary £5,206

# 2-D Projection — PCA & t-SNE

*Clean separation along the first principal component*

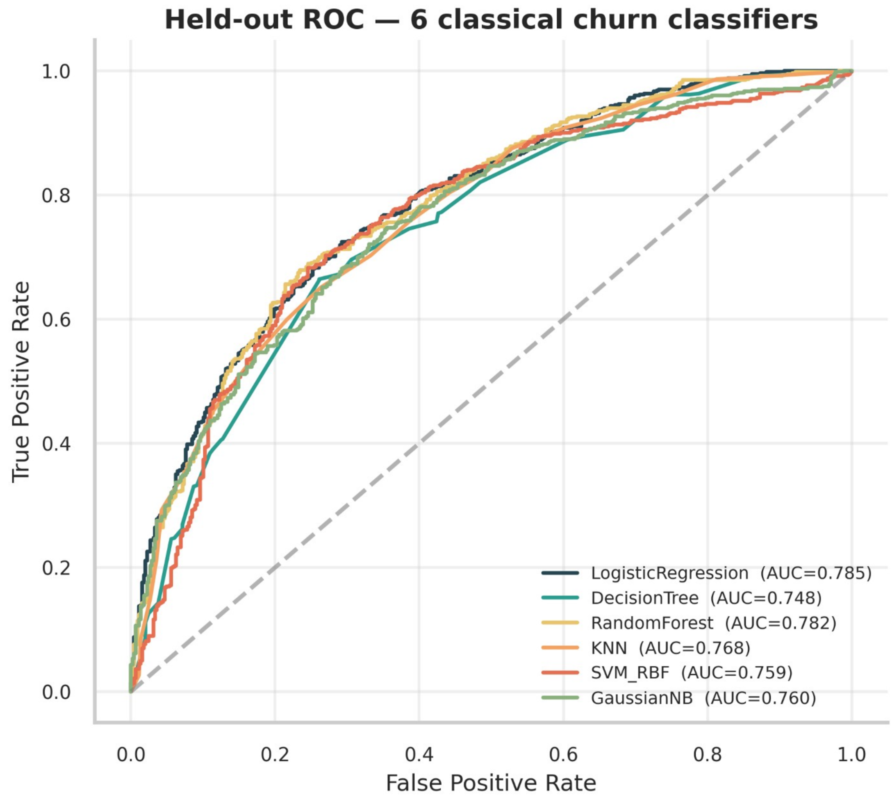


# Churn Classification — 7 Models Compared

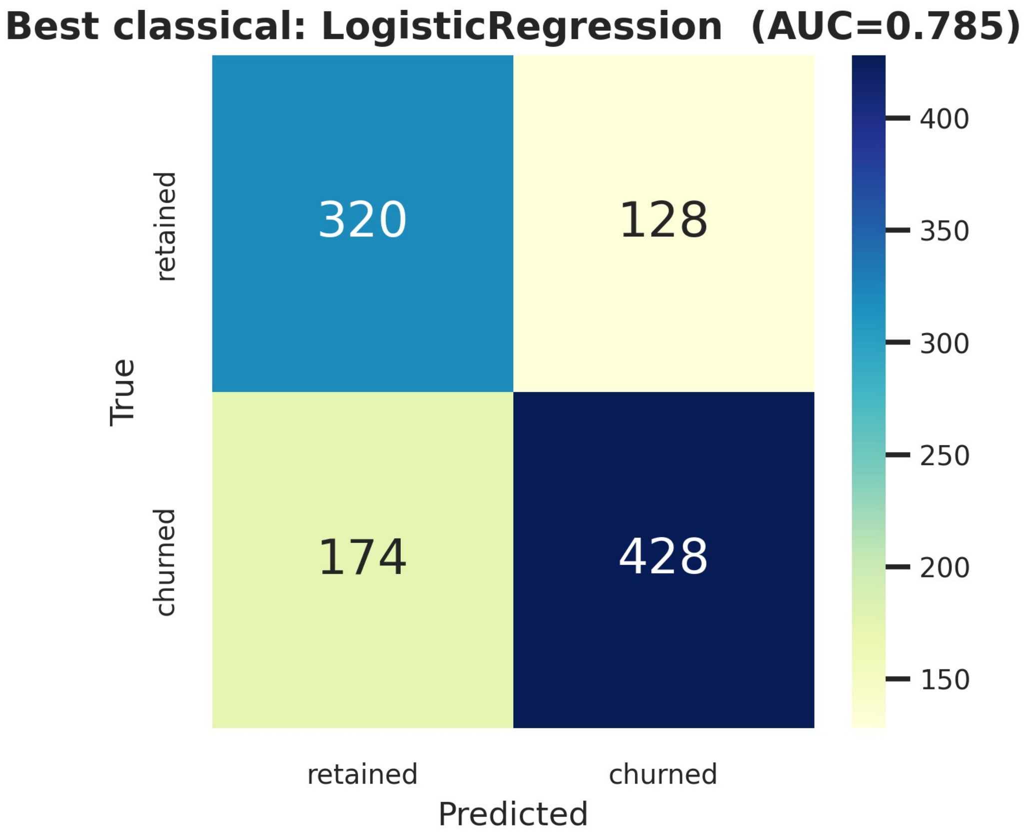
*5-fold CV + held-out test; SMOTE inside folds*

| Model | AUC | F1 | Precision | Recall |
|---|---|---|---|---|
| LogisticRegression | 0.785 | 0.739 | 0.770 | 0.711 |
| RandomForest | 0.782 | 0.745 | 0.747 | 0.744 |
| KNN | 0.767 | 0.721 | 0.740 | 0.703 |
| GaussianNB | 0.760 | 0.744 | 0.724 | 0.766 |
| SVM_RBF | 0.759 | 0.745 | 0.752 | 0.738 |
| DecisionTree | 0.748 | 0.724 | 0.754 | 0.696 |
| MLP (PyTorch) | 0.783 | 0.738 | 0.756 | 0.721 |



# Best Model — LogisticRegression (AUC 0.785)

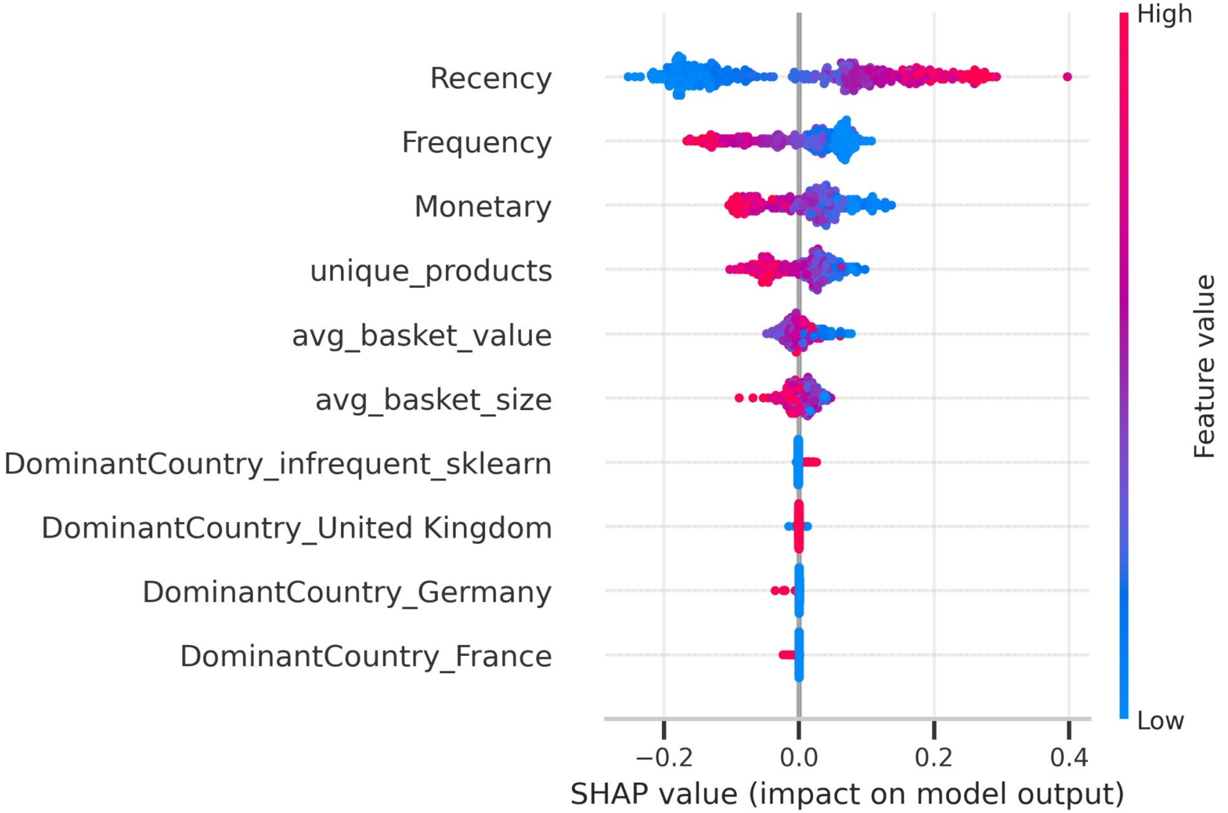
*RandomForest within 0.003 · MLP AUC 0.783*

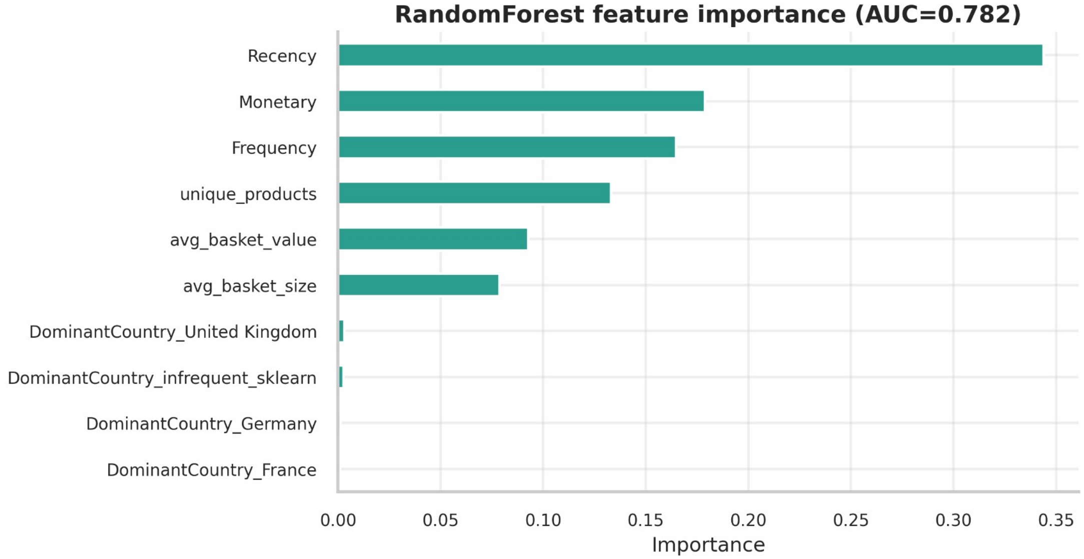



- AUC 0.7850 · F1 0.7392 · Recall 0.7110
- MLP test: AUC 0.7832 · Precision 0.7561 · Recall 0.7209
- Classical and MLP converge — feature ceiling on RFM+behavioural
- → classical models preferred (cheaper, interpretable)
- → DL would need richer signals (sequences, sessions) to pull ahead

# Explainability — SHAP on RandomForest

*Recency dominates; behavioural features add the rest*

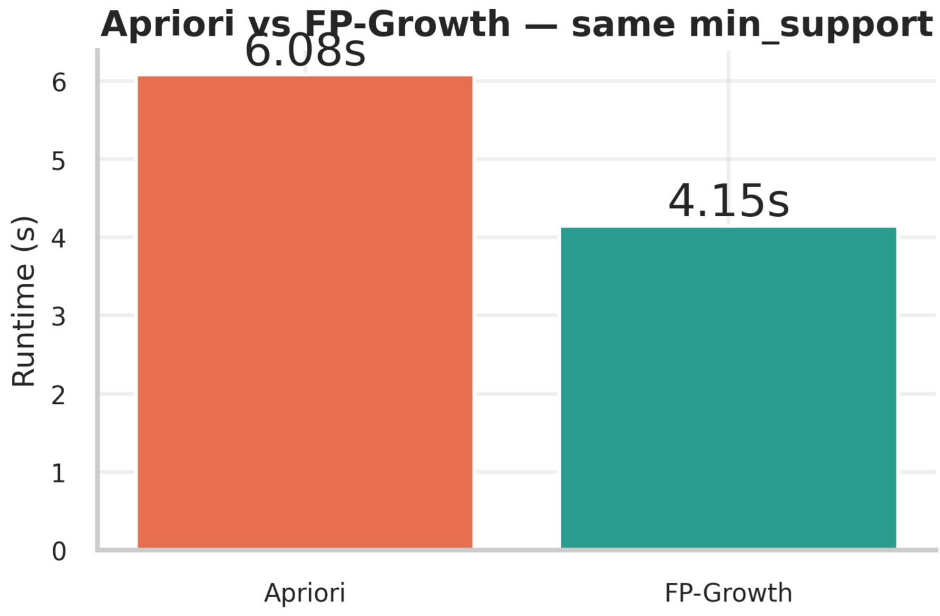


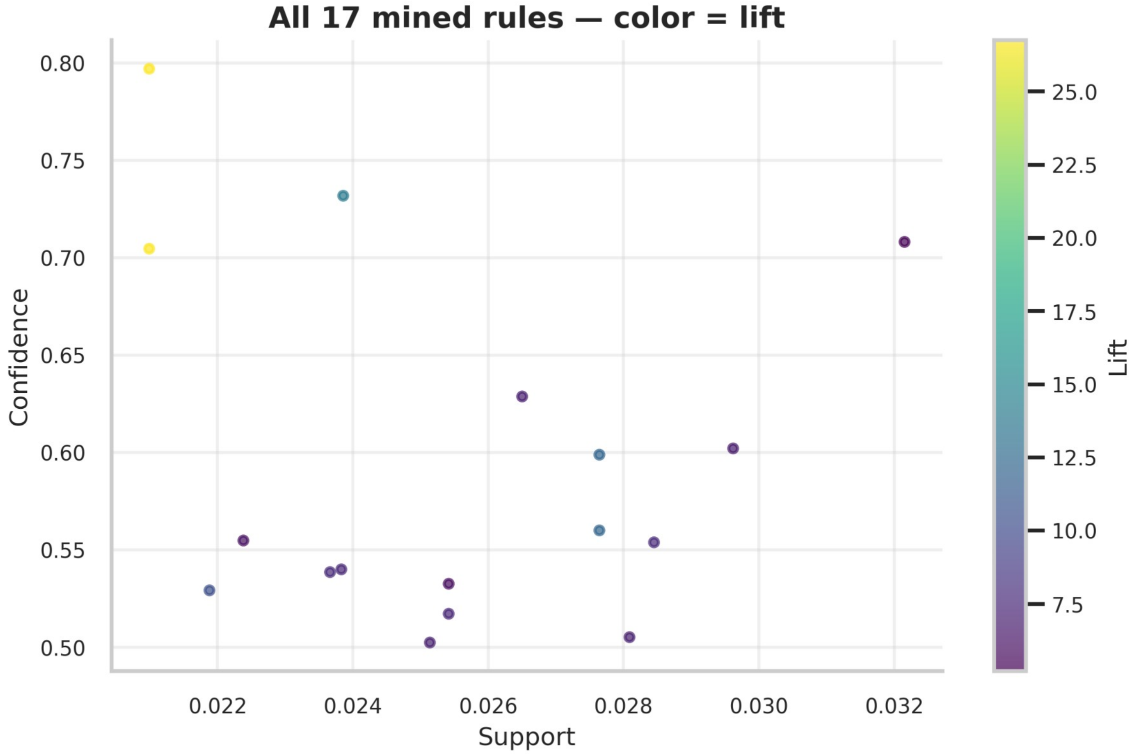



# Market Basket — Apriori vs FP-Growth

*Same 242 itemsets, 17 rules at lift ≥ 1.5 — FP-Growth 1.5× faster*

| Algorithm | Itemsets | Rules | Runtime (s) |
|---|---|---|---|
| Apriori | 242 | 17 | 7.15 |
| FP-Growth | 242 | 17 | 4.25 |



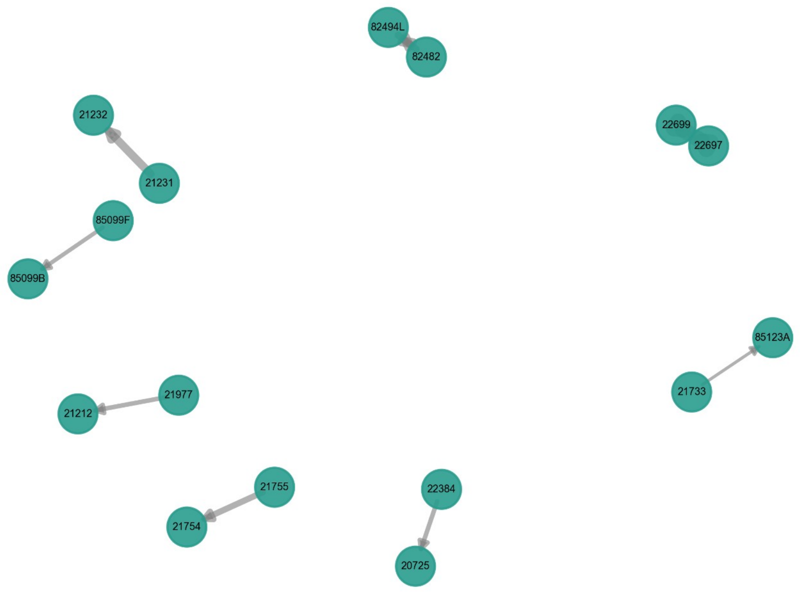



# Top Association Rules

*Strongest signals = collectible sets + color-variant pairs*

| Antecedent | Consequent | Conf. | Lift |
|---|---|---|---|
| ROSES REGENCY TEACUP AND SAUCER | GREEN REGENCY TEACUP AND SAUCER | 0.70 | 26.8 |
| GREEN REGENCY TEACUP AND SAUCER | ROSES REGENCY TEACUP AND SAUCER | 0.80 | 26.8 |
| SWEETHEART CERAMIC TRINKET BOX | STRAWBERRY CERAMIC TRINKET BOX | 0.73 | 13.9 |
| WOODEN PICTURE FRAME WHITE FINISH | WOODEN FRAME ANTIQUE WHITE | 0.60 | 12.1 |
| WOODEN FRAME ANTIQUE WHITE | WOODEN PICTURE FRAME WHITE FINISH | 0.56 | 12.1 |

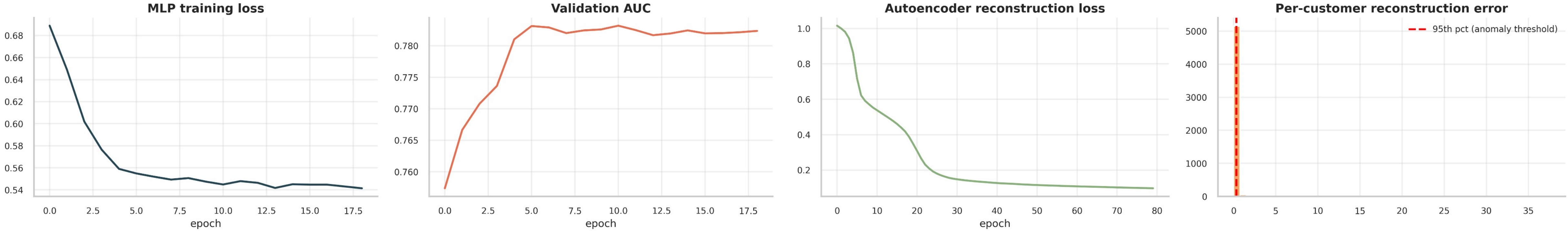
Top-20 association rules (edge width = lift)




Pattern: customers who buy one item from a colour or themed set frequently complete the set.
Operational use: post-purchase recommendations and bundled retention offers.
Rules are mined on customer-attributed invoices only — consistent with the per-segment rules, and excluding anonymous wholesale/bulk baskets.

# Deep Learning Extension

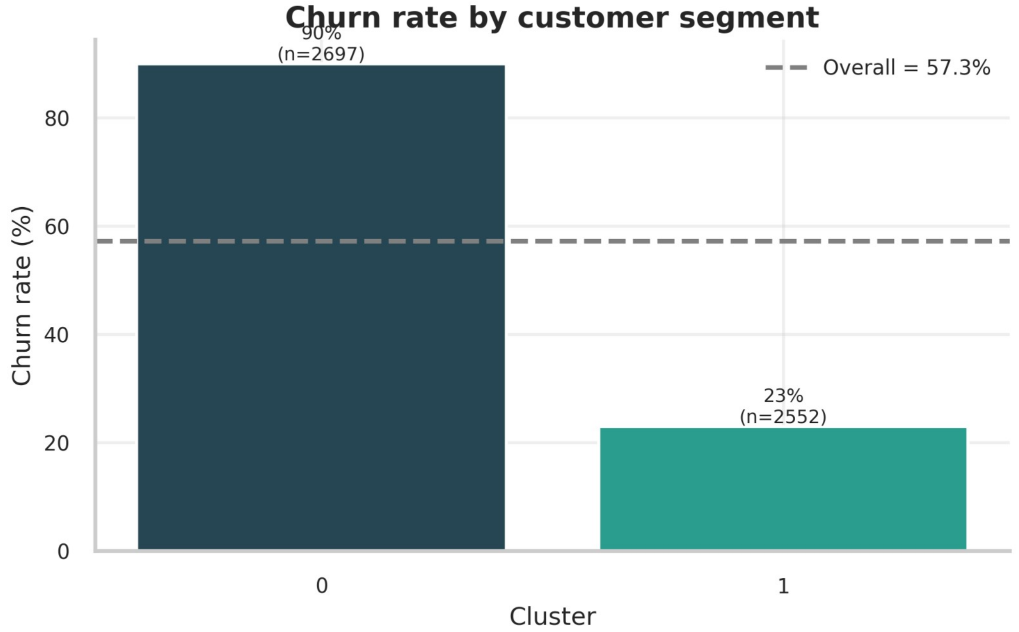
*MLP for churn + Autoencoder for anomaly detection*




MLP [64,32,16] with SMOTE + early stopping reaches test AUC 0.783 (F1 0.738, precision 0.756) — on par with the best classical model. Autoencoder (bottleneck=4) flags the top 5% of customers by reconstruction error as anomalies — candidates for VIP review or fraud screening.

# ★ HEADLINE FINDING — Churn Rate by Segment

*67-percentage-point gap between segments*




| Segment | n | Churn % | Avg Recency | Avg Monetary (£) |
|---|---|---|---|---|
| Dormant | 2,697 | 90.0 | 367.9 | 519.5 |
| Active loyal | 2,552 | 23.0 | 65.2 | 5,401.2 |
| **Overall** | **5,249** | **57.4** | — | — |

**Dormant**

**90%**

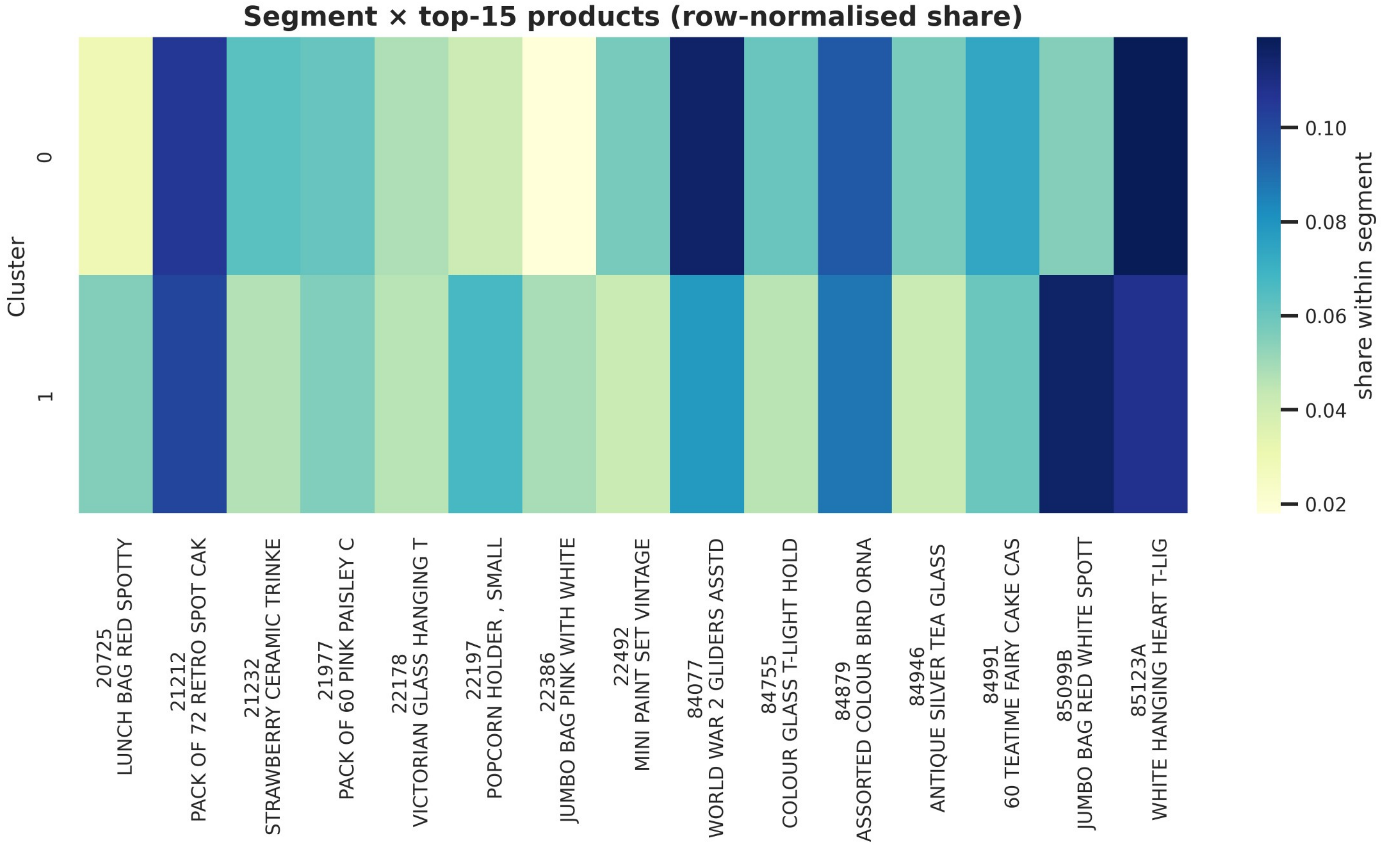
**Active loyal**

**23%**

67-point churn gap
→ segment-aware retention works

# Loyal Segment Buys in Colour Pairs

*Per-segment FP-Growth recovers cross-sell opportunities*




- Pink Lunchbag → Red Lunchbag (lift 7.4)
- Pink Jumbo Bag → Red Jumbo Bag (lift 6.4)
- Red Heart T-light → White Heart T-light (5.3)

**Recommendation: target the 23% churners in the loyal segment with colour-pair bundles. Use win-back campaigns (not bundles) for the dormant majority.**

# Limitations & Future Work

*Honest caveats — and where we'd go next*

## Limitations

- Leakage fix: supervised features built strictly before 2011-09-01 — best AUC now a realistic 0.785 (full-history features were inflated)
- Recency dominant but not sufficient — ablation without it drops AUC to 0.75; baselines: majority 0.50, recency-only 0.75
- Single retailer, UK-centric, 2 yrs — generalisability untested
- No demographic attributes available
- Dormant segment too sparse for in-cluster rules
- Anomaly threshold purely percentile-based

## Future Work

- Transformer on purchase sequences (next-basket + churn jointly)
- Uplift modelling: who responds best to retention offers
- Cross-retailer transfer to test segment-rule stability
- Domain-labelled fraud / VIP ground truth for anomalies

# Conclusion & Thank You

*Q&A — happy to discuss any slide*

## One pipeline, four course topics, one business insight.

- Segmentation: K-means k=2, Silhouette 0.42
- Classification: LogisticRegression AUC 0.785 · MLP AUC 0.783 (leakage-safe pre-cutoff features)
- Association: 17 rules, FP-Growth 1.5× faster, top lift 26.8
- ★ Synthesis: 67-pt churn gap → colour-pair retention bundles

Code, report & slides — Team 4 GitHub repository

Team 4 — Cuong · Danh · Dat · Dinh • UIT, VNU-HCM • 02 June 2026